Learning About People's Attitude Towards Food Available in India and Its Implications for Fair AI-based Systems

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***Abstract*—One of the popular notions of fairness from ethics literature envisages that a human requester of products or services should experience equal access to their choices regardless of their background. With food being essential to human survival and core to any economy, the domain has attracted a number of Artificial Intelligence (AI) techniques applied to tasks like food recommendation, recipe creation and food-oriented wellness. We want to explore if such concerns of fairness are appropriate here since food choices have an impact on an individual's health, personal finance and other aspects of life, and thus, any AI introduced here should be responsibly evaluated for not only performance but also non-functional requirements like their impact on users.**

As a first step towards that goal, in this paper, we want to understand people's perception towards food with respect to their background, and use this as the context to evaluate food-based AI systems in future. Therefore, we conduct the first-of-its-kind survey on perception about food available in India at educational institutes from participants who may associate themselves with different regions within and outside India. We find that users have unequal access to food from different regions today and this may be considered unfair by existing group fairness definitions. Further, we find that many food choices transcend regions, a majority of people prefer food outside their own regions and that usage of digital tools for food is on the rise. The results suggest that AI developers can provide broader and fairer (equal) access by using a mixed strategy of food choices that transcend regions (exploit food data) and also choosing new choices that are region-specific (explore food) dependent on user's regional association.

***Index Terms*—Applied Data Science (DS), Ethical Issues in DS Applications, Food, Survey, AI Implications**

I. Introduction

Artificial Intelligence (AI) services are known to have unstable behavior when subjected to changes in data, models or users. Such behaviors, whether triggered by omission (e.g., neglecting to have a diverse dataset) or commission (e.g., intentional attack), lead to trust issues when AI services work with humans. For example, in the U.S. criminal justice system, Correctional Offender Management Profiling for Alternative Sanctions (COMPAS) system had been used to predict recidivism - the tendency of a convict to re-offend. [1][2] showed that COMPAS was biased against African Americans at twice the rate in comparison to white people. In another case, Amazon AI-based recruitment tools showed bias against women candidates, whose objective was to select the potential candidate's from a bundle of resumes [3].

Food is essential to human survival and core to any economy. Multiple factors motivate a person's choice of food like habits, need, liking, hunger, health, convenience, and pleasure [4] and in turn, food impacts a person's health, personal finance and other aspects of life.

We are interested in studying how AI used in the food domain in a diverse country like India should behave and making them trustworthy. Consider the example of region. A fair AI may be considered one which is equally able to recognize and recommend different food across diverse regions consistently for breakfast, lunch, dinner and snacks. This is in addition to known problems in recommendation systems like favoring popular items (popularity bias) [5], recently ordered items (recency bias), conformity bias, selection bias, exposure bias, and position bias [6].

As a first step towards that goal, in this paper, we design and conduct the first-of-its-kind survey on perception about food available in India at educational institutes from participants who may associate themselves with different regions within and outside India. Our research questions are along two categories - **Category 1: Trends in food and people:** RQ1: Are there foods that transcend regions? RQ2: Do people use digital tools like food recommendation systems? **Category 2: Food preference by regional association** RQ3: Do people feel that the food choices available in India are reflective of what they relate to? RQ4: Are people associating themselves to outside India able to relate to food available in India?

We collected food menu from 15 educational institutions from the northern, southern, eastern and western regions of India covering hundreds of food choices. Of these, we selected data from 12 institutions, three from each region, consisting of 40 food and 40 meal (combo) choices as summarized in Table IV. We then built a web-based survey platform to show online participants from within and outside India food choices randomly for the four regions and recorded their preferences. We analyze these results from the context of AI fairness definitions and draw insights. A total of 124 participants responded to the 40-question survey over a 3 week period drawn from within and outside India to give 4,960 answers. We find that users have unequal access to food from different regions they associate with today and this may be considered unfair by the definition of group fairness, also called statistical parity [7]. Further, we find that many food choices transcend

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regions, a majority of people prefer food outside their own regions and that usage of digital tools for food is on the rise.

The results suggest that AI developers can provide broader and fairer (equal) access by using (exploiting) a mix of choices that transcend regions and also choosing (exploring) new ones that are region-specific.

Thus, our contributions in this work are to (a) understand people's attitudes towards food available in India, (b) discuss its implications in AI-based systems such as recommendation systems and (c) provide an open dataset of food and meal choices for further research in this area. The rest of the paper organizes as follows. In Section II, we discuss the relevant prior work. Section III consists of the survey design, followed by the survey system in Section IV. In Section V, the survey results are described, and Section VI includes the implications of our findings for AI-based food systems. Section VII includes the discussion and conclusion.

II. Related Work

We now discuss the relevant prior work in the areas of technology usage for food services, the definitions for fairness in AI and previous work on fairness in Indian context.

A. Technology Usage in Food Services

Automation is being used while delivering food services keeping the users motivation (demand) into context. In [8], the authors introduced fifteen motivations that people could have while choosing the foods in everyday life: liking, habits, need and hunger, health, convenience, pleasure, traditional eating, natural concerns, sociability, price, visual appeal, weight control, affect regulation, social norms and social image. Another work [4] investigated the relation between motivations and eating occasions, and found that food and beverage decisions are significantly influenced by the different eating occasions at various times of the day. While personal liking is responsible for all eating occasions, function-oriented factors are responsible for day-time eating, and psychological or emotion-oriented needs are responsible for night-time eating. This work [9] dissects the recipes into their culinary elements, e.g., dietary style, ingredients, and nutritional profile details, which helps to make them computable.

B. Fairness in AI Systems

It is a human-value to be fair and lack of fairness is defined as bias. There are many definitions of fairness in literature and a detailed description and illustration on a common data set can be found in [7]. They can broadly be clubbed around the following categories: (a) *individual* definitions which measure whether similar individuals are treated similarly, (b) *group* definitions where statistical property about sub-groups of the population are measured in inputs and outputs, and (c) *others* where a combination or other characteristics of the populations are measured. We note that selection of a suitable bias definition for a domain has socio-technical-legal dimensions and as a start, we select one popular definition for our study.

We will consider the group fairness definition in the paper [7], also called statistical parity or equal acceptance rate. It can be illustrated for a decision, like loan being approved for an applicant (denoted d = 1), with the applicant being of gender (feature) g, which can have values male (m) or female (f), by $p(d = 1|g = m) = p(d = 1|g = f)$. The equation says that a loan approval system is considered fair by this definition if at an aggregate level, applicants associated with male or female gender have equal probability of getting loans approved. The definition assumes that there are sufficiently large number of applicants with the feature of interest (gender here) qualifying for the decisions.

Applying the group fairness definition to our food situation, the decision in favor of a food choice (denoted d=1) is fair with respect to region (r) if the probability of the decision is the same conditioned on user's region value (l_i). Mathematically,

$p(d = 1|r = l_i) = p(d = 1|r = l_j), \forall l_i \neq l_j$

Thus, any system (manual or AI-based recommendation system) is group fair if it offers the consumer equal chance of viewing food choices they like of any region. This assumes that there are sufficiently large food choices in each region of users liking. In applying this definition on a dataset, a practical issue is how to measure the magnitude of difference in probabilities. We will use standard deviation (SD) here - a value of zero indicates all probabilities conditioned on r are same.

C. AI Fairness in Indian Context

Fairness is one of the five AI ethical principles commonly used in literature, others being transparency, non-maleficence, responsibility, and privacy [10]. Kalyanakrishnan et al. described the various opportunities and challenges (linguistic diversity, legacy public records, and healthcare system) for AI in India [11]. They have suggested a need to plan "AI for India" by keeping India's cultural, social, political, and economic configuration in mind.

The authors of [12] talk about algorithmic fairness in India based on their 36 qualitative interviews with researchers and analysis of machine learning (ML) deployments in India. Their findings suggest that there is a need for attention to fair ML in India in three aspects - 1) diversity of data and stability of models to distortions, 2) the risk of double standards that model makers, who are usually from the privileged sections of the society, may have for other sections including marginalized people, and 3) the need to validate AI project aims. Another work [13] discusses the bias in Crime Mapping, Analytics and Predictive System (CMAPS), which Delhi Police announced to use for predictive policing. This work reported the historical bias, representation bias, and measurement bias in CMAPS. We are not aware of any prior work on food and AI fairness.

III. Survey Design

The survey is designed to understand people's perception towards food available in India with respect to their regional background. We now describe our survey questions based on selection of regions, survey structure, questions about the participants, their attitudes towards food choices, and overall trends.

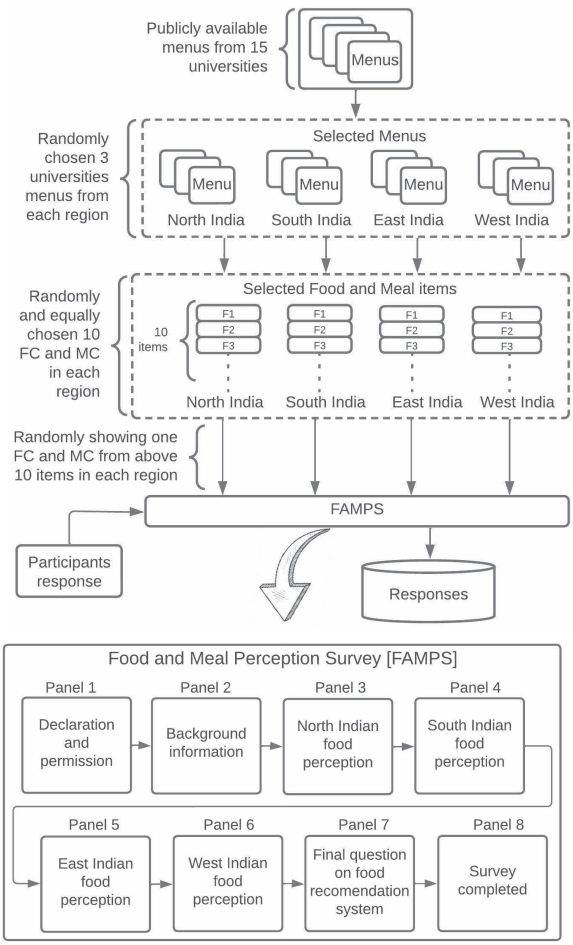


Fig. 1: FAMPS survey (a) flow diagram and (b) panel overview.

A. Defining Regions of India

The choice of region is a critical decision and one can interpret it from different perspectives like political states, socio-cultural regions (including languages) and geographic (including weather, crop cultivation). For our study, we consider a geo-political segregation of India by regions which are Northern Region (NR), Southern Region (SR), Eastern Region (ER), and Western Region (WR). The states and union territories that fall into these regions are found here [14] along with the population as per 2011 census [15]. As we infer from [14] , NR and ER are India's two most populated regions followed by SR and WR.

As explained later, to decide on the respondent's region, we asked them about which region of India (or outside) they related themselves to and to decide which food belongs to which region, we relied on the location (region) of the institute from which the menu was selected. Note that there are other alternatives possible for these decisions and we leave conducting surveys based on them and comparison, for future work.

TABLE I: Bio questions [Panel 2].

Q.No.	Question asked
P2-Q1:	What is your age group?
P2-A1:	18-22 years, 23-44 years, 45-60 years, 61 years and above
P2-Q2:	Which region inside or outside India do you associate with?
P2-A2:	North India, South India, East India, West India, Outside India
P2-Q3:	What is your gender?
P2-A3:	Male, Female, Prefer not to say
P2-Q4:	Do you think your background like gender, type of region, and age has an impact on your food choices?
P2-A4:	Yes, No
P2-Q5:	Which food type category do you belong to?
P2-A5:	Vegetarian, Non-Vegetarian, Vegan, Others
P2-Q6:	Which type of region do you associate with?
P2-A6:	Rural (villages), Semi-urban (towns), Urban, Metros, Others (like college, military cant.)

TABLE II: Food and meal questions [Panels 3, 4, 5 and 6] for NR, ER, SR and WR respectively.

Q.No.	Question asked
P#-Q1:	To what extent have you consumed FC (or a majority of it) in the past week?
P#-A1:	Almost never, Rarely, Sometimes, Often, Quite Often
P#-Q2:	To what extent will you prefer to consume FC next week?
P#-A2:	Almost never, Rarely, Sometimes, Often, Quite Often
P#-Q3:	To what extent do friends around you consume FC in a week?
P#-A3:	Almost never, Rarely, Sometimes, Often, Quite Often
P#-Q4:	To what extent do your family members consume FC in a week?
P#-A4:	Almost never, Rarely, Sometimes, Often, Quite Often
P#-Q5:	If you are given a choice of food item, would you have a preference for FC over any other choice?
P#-A5:	Yes, No, Can't say
P#-Q6:	Suppose you were traveling and had to eat FC for a majority of meal times in a week. How would you feel?
P#-A6:	Would hate it, Okay with it, Would love it
P#-Q7:	Suppose you are setting up the menu of a mess. How likely are you to include this meal selection (shown on right) that includes FC?
P#-A7:	Not likely, Somewhat likely, Likely
P#-Q8:	Suppose you were traveling and had to eat this meals selection for a majority of MC in a week. How would your health be impacted?
P#-A8:	Will get sick, Will not be sick but won't be happy, No impact can do it for another week, Would be much happier and healthier

B. Survey Structure

Our survey consisted of 40 questions asked over 8 panels as described below and shown in Figure 1(b).

1) *Panel 1:* Declaration and taking permission from respondent.
2) *Panel 2:* "Tell us about yourself" asking about the respondent's background information.
3) *Panels 3, 4, 5 and 6:* Randomly chosen Food Choice (FC) and corresponding Meal Choice (MC) shown from NR, SR, ER, and WR respectively. Then asking about

respondent's perception of the choices.

4) *Panel 7:* Thanking respondent and asking about usage of food applications and their preferred food.
5) *Panel 8:* Prompt for completion of survey.

1) About Background of Participants

On second panel, we asked six questions and collected participant's basic background information like their age (P2-Q1), gender (P2-Q2), region they associated with (P2-Q3), participant's thought on impact of background on food choice (P2-Q4). If the response for P2-Q4 was yes, then we asked questions about food preferences (P2-Q5) and sub-regions (P2-Q6). We did not ask any personally identifiable information. The questions and possible responses for characteristics of survey respondents are given in Table I.

2) About Preferences on Food and Regions

On Panels 3 to 6, 8 questions per panel were asked (total 32) to get information about perception of FC and MC shown from NR, SR, ER, and WR respectively. These 8 questions comprised of six questions about FC and two questions on MC, which are listed in Table II along with possible responses for each question.

3) Beyond Presented Choices

Finally, we had two questions to collect information about usage of food applications and participant's preferred food. The questions are listed in Table III.

TABLE III: Questions for more insight [Panel 7].

Q.No.	Question asked
P7-Q1:	Do you use a mobile-app based food recommendation service ("mobile-app service" like Zomato, All-Recipes)?
P7-A1:	Yes, No
P7-Q2:	If there is a particular kind of food that you like to have on a regular basis, do let us know?
P7-A2:	*Text Input*

In our survey, we used the snowball technique to invite the participants through e-mail and social media platforms. Participants were free to share the link with their acquaintances.

IV. Survey System

The Food And Meal Perception Survey system (FAMPS) is designed to show FC and MC to participants and collect data about their perception of it. FAMPS has three key components - data about the FC and MC to be shown, web-based interface to interact with survey participant and an analytic view to report/monitor statistics on survey responses. Its overview is shown in Figure 1.

Here, the data is stored in MySQL database and the web-based interface is developed using PHP, HTML, JavaScript, Bootstrap and CSS. For hosting we used Heroku platform. Based on the data collected, several queries were answered using bar graphs, pie charts, tables, etc., which can be referred from Github [14].

Food data and their regions: The menu was collected from institutes across the four regions of India from publicly available sources. The list can be referred from Github [14]. In terms of terminology, the meal choice available at a given time (breakfast, lunch, or dinner) is called a ***meal***. An example of breakfast meal is: *(Poha + Jalebi + 1 Egg + Cornflakes + Sprouts + ButterMilk + Bread + Bournvita + Tea + Coffee)*, of lunch/dinner is: *(Mix Veg + Chana Dal + Rasmalai + Rice + Dry Ghee Roti + Soaf)*. Each item which can normally be consumed separately is called a ***food***. Full list is found here [14]. Any other item that is not consumed independently is called an ***accompaniment***. Examples are: *(ButterMilk, Bread, Bournvita, Tea, Coffee, Rice, Dry or Ghee Roti, Soaf)*. We have not considered the accompaniments in FC. Overall: (1) *Food choices:* There are 40 FC with 10 chosen for from over 100 available FC for each region as shown in Table IV. (2) *Meal choices:* There are 40 MC each containing corresponding FC, with 10 from each region. (3) The 10 FC and MC taken for each region are from three different universities. (4) The 10 FC and MC taken in each region are approximately equally chosen from breakfast, lunch, and dinner. (5) FC and MC associated with specific regions are found here [14], which are based on the location of the institute from where the data has been taken.

V. Results of the Survey

In this section we will cover the responses received from survey participants, the answers to research questions posed in Section I and other inferences based on collected data.

A. Survey Responses

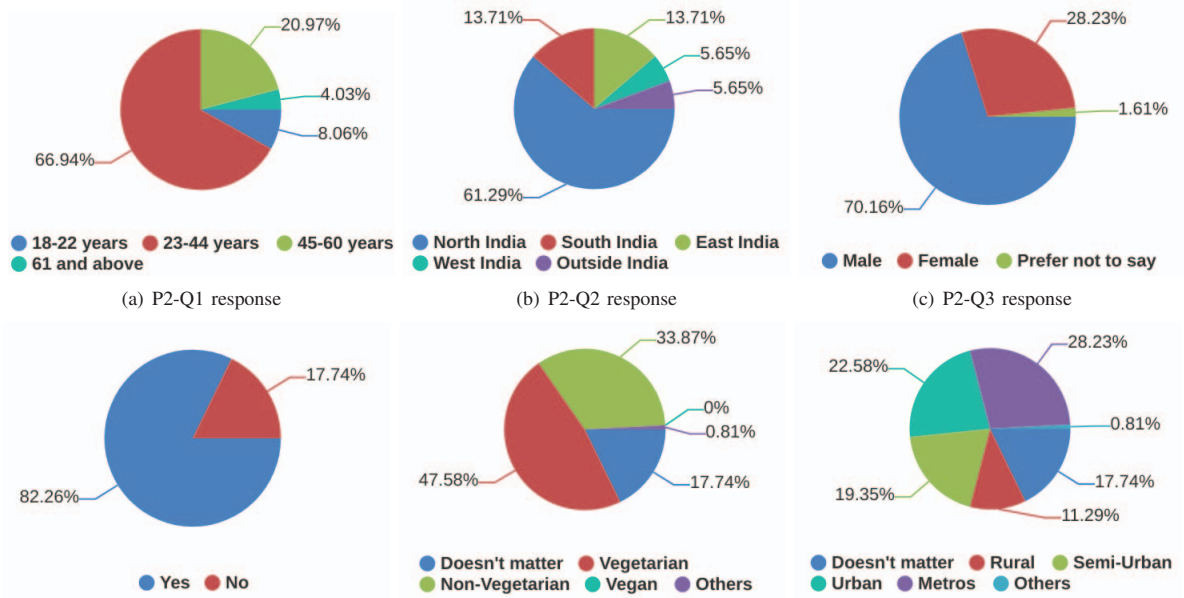
We ran the survey for three weeks from 14 June 2022 to 7 July 2022. With 124 participants responding to 40 questions each using FAMPS, we have a total of 4,960 answers. We now describe the results.

Figure 2 shows the results of the background questions listed in Table I. A majority of the respondents were in the age group of 23-44 (Figure 2(a), 67%), associated with NR region (Figure 2(b), 61%), were male (Figure 2(c), 70%), and thought background impacted their food choices (Figure 2(d), 82%). Majority of those who believed background matters were vegetarian (Figure 2(e), 61%), and were from metros and urban areas (Figure 2(f), 51%).

TABLE IV: Region wise unique food choices based on the location of institute the data originated.

Region	Used FC / Available FC
NR	10/115
SR	10/121
ER	10/100
WR	10/121

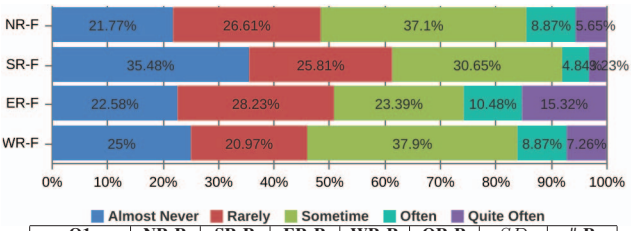
Figures 3 to Figure 10 show the response of P#-Q1 to P#-Q8 listed in Table II. Recall from Section III-B2 that the Panels 3,4,5, and 6 have FC and MC related to NR, SR, ER and WR in a control flow shown in Figure 1(b). Figure 3 summarizes the response of *P#-Q1 - To what extent have you consumed FC (or a majority of it) in the past week?* A majority of people selected *Sometimes* with 37.10% and 37.90% in NR and WR



(a) P2-Q1 response (b) P2-Q2 response (c) P2-Q3 response

(d) P2-Q4 response (e) P2-Q5 response (f) P2-Q6 response

Fig. 2: Respondent background by: (a) age, (b) region, (c) gender, (d) background matters, (e) food eaten, and (f) sub-region.



Q1	NR-R	SR-R	ER-R	WR-R	OR-R	SD	#-R
P3 [NR-F]	**0.184**	0.176	0.0	0.0	0.143	0.0931	0.145
P4 [SR-F]	0.092	0.059	**0.118**	0.0	0.0	0.0534	0.081
P5 [ER-F]	*0.263**	***0.294****	*0.235**	0.143	*0.286**	0.061	*0.258**
P6 [WR-F]	0.171	0.176	0.118	***0.286****	0.0	0.1038	0.161
SD	0.07	0.0959	0.0959	0.1369	0.1369		
P# [#-F]	**0.178**	0.176	0.118	0.107	0.107		0.161

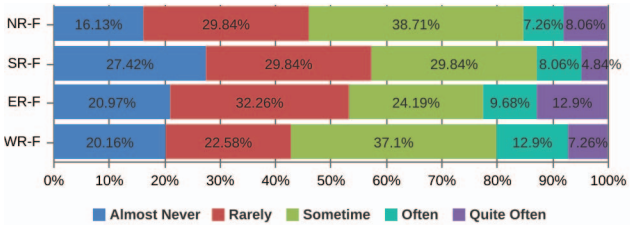
Fig. 3: P#-Q1 responses, SD denote standard deviation.

panels respectively, *Almost Never* with 35.48% in SR panel, and *Rarely* with 28.23% in ER panel.

In the table shown in Figure 3, we provide probabilities of participants' $[p(liking|region)]$ i.e., liking of food conditioned over region of food (suffix **-F**) and region of respondent (suffix **-R**) for *P#-Q1*, where *liking* is defined as participant choosing *Often* or *Quite Often* as their responses. ***#-R*** column represents aggregation over participant's region, similarly ***#-F*** row represents aggregation over food's region. Numbers in bold represent the best values across rows, and those with an asterisk (*) represent the best values across columns, except for the last two rows and columns, which are derived statistics. Ideally, for group fairness, the probability of liking FC from any region by specific region's participant (means across the column) should be equal and also the probability of liking specific region's FC by any region's participant should be equal (means across the rows) [Figure 3]. However, we observe variation in the values due to which standard deviation is non zero. This indicates biasness by group fairness definition. Moreover, we observe that ER-F is broadly favored over other foods by participants from different regions, except WR-R.

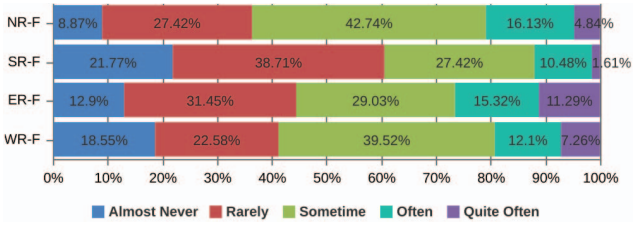
Similarly, Figure 4 summarizes the response of *P#-Q2 - To what extent will you prefer to consume FC next week?* The majority of people selected *Sometimes* in NR, SR, and WR panels with 38.71%,29.84%, and 37.1% respectively, while 32.26% of people selected *Rarely* in ER panel. Also, we notice that ER-F is favored by all participants in P#-Q2. To gauge the perception of people in the social circle of the participant, we asked *P#-Q3 - To what extent do friends around you consume FC in a week?* Figure 5 summarizes the response where the majority of people in NR and WR panels opted for *Sometimes* with 42.74% and 39.52% respectively, and the majority of people in SR and ER panels selected *Rarely* with 38.71% and 31.45%, respectively. And we notice that for P#-Q3, ER-F is favored by all participants. We asked *P#-Q4 - To what extent do your family members consume FC in a week?*, and Figure 6 summarizes its response. The majority of people from NR and WR panels selected *Sometimes* with 37.1% and 36.29% respectively. In ER and SR panels, the majority of people selected *Rarely* with 31.45% and 36.29%. Also, ER-F and WR-F are favoured by equal number of regions when asked P#-Q4.

To know if the participant will prefer the FC willingly, we asked in *P#-Q5 - If you are given a choice of a food item, would you have a preference for FC over any other choice?* From Figure 7 it is interesting that a majority of people from ER and WR panels opted for *Yes* and majority of people from NR and SR panels selected *No*. We observe that WR-F is most preferred by all participants (except SR-R) in Figure 7. Now, to know if the participant would like to eat FC shown, we asked in *P#-Q6 - Suppose you were traveling and had to eat FC*



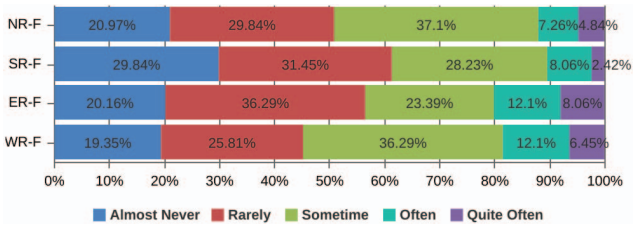
Q2	NR-R	SR-R	ER-R	WR-R	OR-R	SD	#-R
P3[NR-F]	0.171	**0.294**	0.0	0.0	*0.143**	0.1247	0.153
P4[SR-F]	**0.171**	0.059	0.118	0.0	0.0	0.0749	0.129
P5[ER-F]	*0.211**	***0.294****	***0.294****	*0.143**	*0.143**	0.0756	*0.226**
P6[WR-F]	*0.211**	**0.235**	0.176	*0.143**	*0.143**	0.041	0.202
SD	0.0231	0.1112	0.1223	0.0826	0.0715		
P# [#-F]	0.191	**0.221**	0.147	0.071	0.107		0.177

Fig. 4: P#-Q2 responses, SD denote standard deviation.



Q3	NR-R	SR-R	ER-R	WR-R	OR-R	SD	#-R
P3 [NR-F]	0.211	***0.353****	0.176	0.0	*0.143**	0.1247	0.21
P4 [SR-F]	0.132	0.118	0.059	**0.143**	***0.143****	0.0351	0.121
P5 [ER-F]	*0.263**	***0.353****	*0.235**	*0.286**	*0.143**	0.0768	*0.266**
P6 [WR-F]	0.211	0.176	***0.235****	0.143	0.0	0.0924	0.194
SD	0.054	0.1213	0.083	0.1168	0.0715		
P# [#-F]	0.204	**0.25**	0.176	0.143	0.107		0.198

Fig. 5: P#-Q3 responses, SD denote standard deviation.



Q4	NR-R	SR-R	ER-R	WR-R	OR-R	SD	#-R
P3 [NR-F]	0.132	***0.235****	0.0	0.0	0.143	0.1013	0.121
P4 [SR-F]	0.118	0.0	**0.176**	0.143	0.0	0.0824	0.105
P5 [ER-F]	*0.211**	0.118	*0.235**	0.143	***0.286****	0.0684	*0.202**
P6[WR-F]	0.171	*0.235**	*0.235**	***0.286****	0.0	0.1114	0.185
SD	0.0418	0.1125	0.1112	0.1168	0.1369		
P# [#-F]	0.158	0.147	**0.162**	0.143	0.107		0.153

Fig. 6: P#-Q4 responses, SD denote standard deviation.

for a majority of meal times in a week. How would you feel? Figure 8 shows that the majority of people from all panels selected *Okay With It*, which shows the tolerant nature of the participants. Also, less participants are likely to go for WR-F which means people are more tolerant towards WR food.

Similar to the table for Q1 in Figure 3, we also show probabilities of participants' *liking* conditioned over region in Figures 4, 5, 6, 7, 8, 9, and 10 for questions P#-Q2, P#-Q3, P#-Q4, P#-Q5, P#-Q6, P#-Q7, and P#-Q8, respectively. Where *liking* is defined as participant choosing *Often* or *Quite Often* for P#-Q2, P#-Q3, P#-Q4 and choosing *Yes, Hate It, Not Likely* for P#-Q5, P#-Q6, P#-Q7 respectively and *Will Be Sick* or *Not Happy* for P#-Q8 as their responses. The underlined numbers in the Tables of Figures 8, 9, and 10 reinforce that low value is better. We again see a non-zero standard deviation indicating biasness by group fairness definition. If we look at the responses for P#-Q1 to P#-Q6 together, we observe that ER-F and WR-F is broadly favored over other foods by participants broadly.

To know if the participant will prefer the MC (containing the above FC) willingly, we asked *P#-Q7 - Suppose you are setting up the menu of a mess. How likely are you to include this meal selection (shown on right) that includes FC?* Figure 9 summarizes the response of it. And majority of people from all regions selected *Somewhat likely*, which shows the tolerant nature of participants. And, to know if the participant will like if they had to eat MC shown we asked in *P#-Q8 - Suppose you were traveling and had to eat this meals selection for a majority of MC in a week. How would your health be impacted?* Figure 10 shows minority of people from all regions selected *Quite Often*, while majority of people from NR and WR panels selected *Sometimes* and majority of people from SR and ER panels selected *Rarely*.

To know the usage of food applications/systems we posed a question *P7-Q1 - Do you use a mobile-app based food recommendation service ("mobile-app service" like Zomato, AllRecipes)?* We found 63.71% responded with *Yes* while the rest 36.3% responded with *No*. Also, majority of the participants responded with *Yes* belong from urban and metros which is 66.15%, and most people saying *No* are from semi-urban which is 32.43%. In order to get an idea about what food items would participant prefer other than what was shown to them, we asked *P7-Q2* if there was a particular kind of food that they would like to have on a regular basis for which we received food items and categories of food items as responses like Green vegetables, Lady finger, Lentils, and many more, full list of responses for this question is available on Github [14].

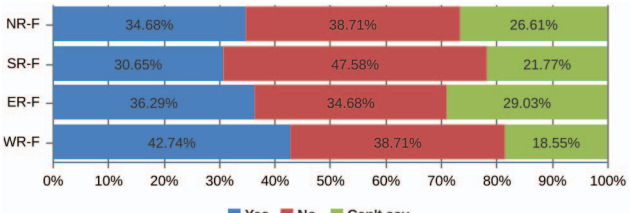
B. Result Summary

Category 1: Trends in food and people

RQ1 - Are there foods that transcend regions?

This question can be interpreted as food that are widely available and food that is widely liked by people. For the former, we found that 35% out of 40 FC were in only one region, 10% in two, 15% in three, and 40% were available in all four regions.

For the latter, we have found that there are food items that are liked across regions. Food items like *Dal Arahar, Aloo-gobhi-curry, Butter-Chicken, Veg chowmein, Aloo carrot peas sabji, Malai Kofta, Lauki Kofta, Egg Curry, and Beans/Snakegourd Thoran(Coconut)* were liked by people from four and above regions out of five regions (NR, SR, ER, WR, and OR). It may be due to increased awareness of food from different parts of the country in educational settings from where we obtained our data and most of the participants may have been from, and also due to people willing to explore new choices.



Q5	NR-R	SR-R	ER-R	WR-R	OR-R	SD	#-R
P3[NR-F]	***0.421****	*0.353**	0.118	0.143	0.286	0.1313	0.347
P4[SR-F]	**0.355**	*0.353**	0.176	0.286	0.0	0.1497	0.306
P5[ER-F]	0.382	0.294	0.353	0.143	**0.571**	0.1548	0.363
P6[WR-F]	*0.421**	0.294	*0.471**	*0.429**	***0.714****	0.1537	*0.427**
SD	0.0323	0.0341	0.1621	0.1369	0.3166		
P# [#-F]	**0.395**	0.324	0.279	0.25	0.393		0.361

Fig. 7: P#-Q5 responses, SD denote standard deviation.

RQ2 - Do people use digital tools like food recommendation systems? In our study, we found that most people saying *Yes* to the usage of the food recommendation system are from urban and metros region (66.15%), and most people saying *No* is from the semi-urban region (2.43%). One of the reasons for having that in urban and metros, more food delivery services are available in comparison to semi-urban. Overall, approximately 64% of participants used online mobile applications like zomato, etc., to recommend the food. It is more than twice the number of people who don't use any food recommender system. It shows that usage of digital tools is picking up.

Food preference by regional association

RQ3 - "Do people feel that the food choices available in India are reflective of what they relate to"?

In our study, we found that 42.11% of NR-R liked FC from NR, 32.89% did not like it, and 25% were unsure about it. SR-R who liked, disliked, and could not say about SR-F were 32.29%, 29.41%, and 35.29%, respectively. The same percentage of ER-R liked and disliked FC from ER, that is 35.29%, while the rest 29.41% opted for *Can't say*. While 42.86% of WR-R liked FC from WR, 57.14% did not. The overall majority (40.17%) of participants felt that food choices in India reflect what they relate to in their corresponding regions, while 34.19% did not feel the same and 25.64% were unsure about it.

RQ4 - Are people associating themselves to outside India able to relate to food available in India?

From such participants, 42.86% of them liked the shown Indian food items, whereas 39.29% disliked the food items and the remaining 17.86% of people opting for *Can't say*. This shows that a majority of people outside India ($OR - R$) were able to associate with foods in India. From Tables in Figures 3-8, we see that $OR - R$ has stronger preference for food from some regions (like $ER - F$) over others.

VI. Implications of Findings for AI based Food Systems

AI methods are increasingly being used in the food domain as shown by literature and our own survey. In this section, we discuss how our survey results have implications for typical AI being developed, especially in India. For exposition, we will consider the example of a hypothetical food recommendation app, *Yumato*, that can suggest meal options to a user along with an eatery where the food may be available, key ingredients and price.

AI relies on data for training its models. Although we have collected over 100+ food and corresponding meal choices per region from the menus, we only used a fraction - 10 FC and 10 MC - in order to have a balanced and non-sparse participant response set. The dataset can be further augmented by associating the food with their images, ingredients and preparation steps to make them easily discoverable by users beyond just names.

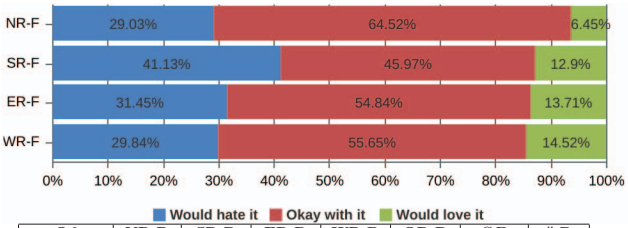
Another implication of the study is on associating food to a particular region. We find that the same food may be widely available across regions (4 food regions) and it may also be preferred by respondents from different regions (5 respondent regions). Information about foods that are at the sweet spot of high availability and preference are valuable to *Yumato* or any food service provider to exploit. In particular, at the beginning of interaction with a user about whom a recommender does not have any data, it can use such food information as defaults. The information about food that is not widely available is also valuable because they provide opportunities for *Yumato* to explore people's preference. But how does the recommender mix the two strategies of exploiting and exploring? This is the typical learning problem studied in sequential decision making using methods like reinforcement learning [16]. Further, our study shows that respondents in India are tolerant of food choices from other regions and thus, diversity may not only be a matter of increasing food access but also good business to keep users engaged with a food platform like *Yumato*.

One can also use the results for making food choices more accessible to specific segments of users like people outside India which one may want to focus on to promote tourism and regional economy. One can also use it for promoting food from specific regions of the country to create local demand, build sustainable food supply chains and generate employment. A long-term aim of knowledge about food and people's preferences based on regions would be to recommend healthier food options to people to manage lifestyle diseases like diabetes or cardiovascular conditions [17].

VII. Discussion and Conclusion

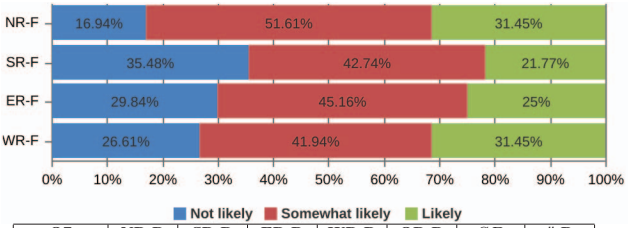
We now discuss the limitations of the current work, opportunities for the future, and put our contributions in perspective.

This study has several limitations. First, FAMPS is a web-based application that uses the English language but English is not the first language of most residents of India. Second, in the FAMPS tool, we asked questions based only on the names of food choices, but other aspects like ingredients involved, preparation steps, images, and nutrient values were not shown to the user. Hence each respondent might answer the questions based on their imagination of the FC shown. Third, we have associated a FC to the region based on where the University from which we obtained the menu was located. As noted, we found the analysis of menus that some food can be trans-region but we did not consider it and assigned it to a single



Q6	NR-R	SR-R	ER-R	WR-R	OR-R	SD	#-R
P3[NR-F]	***0.25****	0.353	0.294	*0.286**	0.571	0.1285	*0.29**
P4[SR-F]	0.342	**0.294**	0.529	0.714	0.857	0.24	0.411
P5[ER-F]	0.316	0.353	0.294	0.429	***0.143****	0.1051	0.315
P6[WR-F]	0.329	***0.235****	***0.235****	*0.286**	0.286	0.0399	0.464
SD	0.0409	0.0565	0.1303	0.2018	0.3166		
P#[#-F]	**0.309**	**0.309**	0.338	0.429	0.464		0.329

Fig. 8: P#-Q6 responses, SD denote standard deviation.

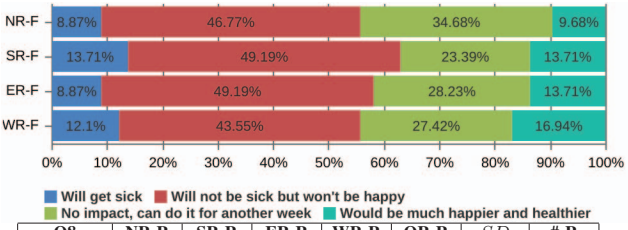


Q7	NR-R	SR-R	ER-R	WR-R	OR-R	SD	#-R
P3[NR-F]	***0.092****	0.353	*0.235**	0.286	*0.286**	0.098	*0.169**
P4[SR-F]	0.368	0.353	0.294	***0.143****	0.571	0.1542	0.355
P5[ER-F]	0.289	*0.235**	0.353	0.429	***0.286****	0.0747	0.298
P6[WR-F]	**0.25**	0.294	0.294	0.286	*0.286**	0.0183	0.266
SD	0.1161	0.0565	0.0482	0.1168	0.1425		
P# [#-F]	**0.25**	0.309	0.294	0.286	0.357		0.272

Fig. 9: P#-Q7 responses, SD denote standard deviation.

region in the survey design for simplicity of exposition. One can conduct advanced analysis to partially associate food to multiple regions, e.g., frequency/ support analysis of food by region. Fourth, this survey was conducted in summer while the respondents may have different preferences with seasons. One can extend our work by addressing these limitations as well as conducting wider surveys with more food and meal choices, and number of participants.

To conclude, this work investigated the people's perception about food in India through web-based questionnaire tools-FAMPS. We found that users have unequal access to food from different regions they associate with today and this may be considered a form of unfairness. Further, we found that many food choices transcend regions, a majority of people prefer food outside their own regions and that usage of digital tools for food is on the rise. The results suggest that AI developers can provide broader and fairer (equal) access by using (exploiting) a mix of choices that transcend regions and also choosing (exploring) new ones that are region-specific.



Q8	NR-R	SR-R	ER-R	WR-R	OR-R	SD	#-R
P3[NR-F]	*0.513**	***0.471****	0.647	0.714	0.857	0.156	*0.556**
P4[SR-F]	0.618	0.588	0.647	***0.571****	0.857	0.1159	0.629
P5[ER-F]	0.618	0.588	0.471	0.714	***0.286****	0.1642	0.581
P6[WR-F]	0.566	0.647	***0.412****	0.714	0.429	0.1325	*0.556**
SD	0.0502	0.0738	0.1211	0.0715	0.2942		
P#[#-F]	0.579	0.574	**0.544**	0.679	0.607		0.581

Fig. 10: P#-Q8 responses, SD denote standard deviation.

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